

# Identifying core intervention components for an AI-XR-based telehealth platform (Attachment-BRIDGE) to support families of preterm infants: A two-round modified delphi study

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## Abstract

**Objective:** Families of preterm infants experience significant emotional and caregiving challenges during neonatal intensive care unit (NICU)-to-home transition, with post-discharge decline in structured professional support. In this study, we aimed to identify core intervention components for an artificial intelligence (AI)-extended reality (XR)-based telehealth platform (Attachment-BRIDGE) to support families of preterm infants during the NICU-to-home transition. **Methods:** Using a Delphi consensus approach, a two-round study with a multidisciplinary panel of 22 participants (NICU nurses, neonatologists, paediatric nursing faculty, AI/XR engineers, and parents of preterm infants) was conducted in the Republic of Korea (1 July–29 August 2025). In Round 1, potential intervention components across three phases—early NICU admission, ongoing NICU hospitalisation, and post-discharge transition—were generated through open-ended responses and literature review. In Round 2, components were rated for importance and performance to identify priority-intervention needs. **Results:** We identified key intervention components that were consistently prioritised across phases, particularly related to emotional stabilisation, attachment-focused interaction, XR-guided caregiving practices, immersive emotional attunement, and simulation-based emergency preparedness. The largest priority gaps occurred during the mid-NICU hospitalisation phase of intensified parental involvement and caregiving responsibilities. **Conclusion:** We delineated a three-stage intervention framework—emotional stabilisation, attachment-focused interaction, and care preparedness—to design an AI-XR-based telehealth platform supporting families of preterm infants across the NICU-to-home transition.

## Keywords

preterm infants, family-centred care, telehealth, artificial intelligence, AI, extended reality, XR, delphi technique

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## Introduction

Worldwide, the total fertility rate—once regarded as a demographic safeguard at the average level of the Organization for Economic Co-operation and Development (OECD)—has declined from 3.3 in 1960 to an all-time low of 1.51 in 2022,

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signalling an unprecedented phase of demographic change.<sup>1</sup> This worldwide trend toward low fertility extends beyond a simple reduction in birth numbers and is closely associated with delayed childbearing and an increasing proportion of pregnancies among women of advanced maternal age ( $\geq 35$  years). Although advanced maternal age in itself may not directly cause adverse neonatal outcomes, delayed childbearing is frequently accompanied by obstetric risk factors, such as hypertensive disorders of pregnancy and prior caesarean delivery, that collectively contribute to a higher prevalence of medically high-risk births, including preterm and low-birth weight infants requiring specialised neonatal care.<sup>2</sup> Consequently, low fertility has emerged as a critical public health challenge with long-term implications for neonatal outcomes, family well-being, and healthcare systems.<sup>3</sup>

Within this broader international context, the Republic of Korea represents one of the most extreme cases globally, and has consistently reported the lowest total fertility rate among the OECD countries. In 2023, the decline in the total fertility rate to 0.72, which is less than half of the OECD average, indicated a rapidly intensifying demographic crisis.<sup>4</sup> Despite this decline in overall births, the proportion of infants born before 37 weeks of gestation reached 10.2%. This represents an approximately 1.5-fold increase compared with a decade earlier and places the country at the upper end of the global preterm birth range (4%–16%), which averages approximately 10% worldwide.<sup>5</sup> In parallel, multiple births increased from 3.5% in 2014 to 5.7% in 2024, with the highest prevalence observed among mothers in their late 30s.<sup>4</sup> These trends indicate that declining fertility does not necessarily translate into reduced neonatal care needs; rather, it coincides with a growing relative burden of preterm and medically high-risk infants.

Owing to physiological immaturity, preterm infants frequently require prolonged hospitalisation and continuous monitoring in neonatal intensive care units (NICUs); thereafter, families must assume complex caregiving responsibilities with limited access to structured professional support.<sup>5,6</sup> The transition from NICU to home requires parents to manage not only the infant's physical health but also their long-term developmental needs, often under conditions of heightened psychological stress and social isolation.<sup>6</sup> During NICU hospitalisation, parents commonly experience restricted physical contact and limited opportunities to participate in their infant's care, which heightens psychological distress, reduces parenting self-efficacy, and complicates early parent–infant attachment. Concurrently, preterm and low-birth weight infants face elevated risks of rehospitalisation, developmental delay, and long-term health vulnerability,<sup>7</sup> which contribute to persistent gaps during the transition from hospital to home.<sup>8</sup>

Family-centred care (FCC) has been widely adopted as a guiding framework to address these challenges. FCC regards families as essential partners in care planning and delivery and emphasises respect, communication, and active participation.<sup>9,10</sup> Despite the broad international consensus on the core principles of FCC, the scope of who is considered “family” and the manner in which FCC is operationalised vary across cultural and healthcare contexts.<sup>11</sup> In global healthcare standards, including those aligned with the Institute for Patient- and Family-Centered Care (IPFCC), FCC extends beyond the nuclear family to include extended family members and other self-identified support persons as integral members of the care team.<sup>12,13</sup>

In contrast, FCC implementation in the Republic of Korea remains limited in practice and tends to prioritise parents as the primary participants, rather than emphasising broader family involvement.<sup>10,14</sup> Institutional constraints, such as high-density hospital environments, infection control policies, and structural limitations of NICUs, have restricted caregiving participation largely to parents.<sup>10,14</sup> Furthermore, Korean studies consistently report a gap between high recognition of FCC principles and limited practical implementation, citing shortages of staffing, time, formal FCC systems, and institutional support as persistent barriers.<sup>14,15</sup> Within this context, telehealth and e-health interventions have emerged as complementary strategies to support caregivers by reducing parental stress, enhancing readiness for home care, and improving caregiving self-efficacy among families of preterm infants.<sup>16,17</sup>

Recent advances in digital health offer opportunities to address limitations in caregiver support during the NICU-to-home transition. Telehealth and e-health interventions have demonstrated benefits in reducing parental stress, enhancing readiness for home care, and improving caregiving self-efficacy among caregivers (parents) of preterm infants.<sup>16,17</sup> Building on these approaches, emerging artificial intelligence (AI) and extended reality (XR) technologies enable immersive simulation and more personalised educational support, with potential to strengthen continuity of care beyond NICU hospitalisation.<sup>18,19</sup>

While human-delivered approaches, including home visiting programs, community health workers, and peer support interventions, offer valuable caregiver support, their scalability and post-discharge continuity are constrained by staffing shortages, restricted caregiver access during NICU hospitalisation, and limited institutional infrastructure for follow-up care in the Korean context.<sup>10,14</sup> In contrast, AI–XR-based telehealth platforms offer complementary structural advantages, including continuous accessibility, immersive simulation-based learning, and adaptive personalised feedback that can support caregivers across the entire NICU-to-home trajectory.<sup>16–19</sup> Furthermore, the multidisciplinary composition of the Delphi panel—encompassing AI/XR technology specialists alongside NICU nurses, clinicians and caregiver representatives—enabled evaluation not only of content priorities but also of the feasibility and appropriateness of digital delivery within this population.

However, existing studies have largely examined these digital modalities in isolation or at discrete timepoints, and the literature lacks a systematic, consensus-based identification of core components for AI–XR-based telehealth interventions tailored to caregivers (parents) during the NICU-to-home transition. Therefore, this study aimed to identify essential intervention components of an AI–XR-based telehealth platform (Attachment-BRIDGE) using a multidisciplinary Delphi approach to establish an evidence-based framework for longitudinal caregiver support beyond hospitalisation.

## Methods

### *Study design*

This study employed a two-round modified Delphi consensus approach to identify the core intervention components of an AI–XR-based telehealth platform (Attachment-BRIDGE; A-BRIDGE) intended to support families of preterm infants during the transition from the NICU to home. The Delphi method enables the systematic synthesis of expert judgment through iterative rounds of structured feedback and is therefore well suited for concept development in such contexts.<sup>20</sup> Data collection was conducted in two rounds in the Republic of Korea: Round 1 (1–21 July 2025) and Round 2 (25 July–29 August 2025). This study was reported in accordance with the DELPHISTAR (Delphi studies in social and health sciences—Recommendations for an Interdisciplinary Standardized Reporting) reporting guideline.<sup>21</sup>

### *Participants*

A multidisciplinary panel was constructed to ensure diverse and practice-relevant perspectives related to NICU transitional care. The panel comprised 20 experts, including NICU nurses ( $n = 7$ ), neonatologists or paediatric faculty specialising in neonatology ( $n = 4$ ), paediatric nursing faculty ( $n = 2$ ), and technology experts specialising in artificial intelligence and extended reality ( $n = 7$ ). In addition, two caregivers, specifically one mother and one father of preterm infants with direct experience of NICU hospitalisation and subsequent discharge, were included as user representatives. Both caregivers participated in all Delphi rounds and contributed equally to the consensus process. Their sociodemographic characteristics and their infants' clinical profiles are summarised in [Table 1](#).

Methodological guidelines recommend a Delphi panel size of 10–30 participants to ensure adequate expertise and stable consensus.<sup>22,23</sup> Considering the interdisciplinary nature and complexity of developing an AI–XR-based neonatal telehealth platform, a target sample of 20–22 participants was established. Ultimately, 20 experts and 2 caregivers completed all Delphi rounds, yielding a final panel of 22 participants, which was deemed sufficient based on prior Korean Delphi studies demonstrating reliable consensus with panels of 10–15 experts.<sup>24,25</sup> Detailed eligibility and exclusion criteria for each participant group are provided in [Supplementary Table S1 \(Supplementary File S1\)](#).

### *Delphi procedures and survey materials*

Data collection was conducted in sequential stages. Round 1 (open-ended survey) was carried out from 1–21 July 2025; Round 2 (structured Delphi survey) was conducted from 25 July to 26 August 2025. When necessary, follow-up one-to-one expert consultations were conducted on 29 August 2025. All study procedures were performed in the Republic of Korea. The overall procedure and timeline of the study are summarised in [Supplementary Figure S1 \(Supplementary File S2\)](#). In Round 1, an exploratory qualitative inquiry was conducted to elicit expert and caregiver perspectives on intervention needs across the NICU-to-home transition, and responses were analysed using content analysis to generate preliminary items and phase-based categories. In Round 2, a structured survey was administered to evaluate the importance, current performance, and future expectations of the identified components. Detailed descriptions of the Delphi procedures, survey materials, and qualitative analysis processes are provided in the Supplementary Materials ([Supplementary File S2](#)). The full Round 1 interview guide and Round 2 Delphi questionnaire are provided in [Supplementary File S5](#).

The survey instrument for Round 2 was developed through a two-step process to ensure content relevance and structural integrity. First, potential intervention components were extracted from the qualitative content analysis of Round 1 responses and a systematic literature review focusing on NICU-to-home transitional care. Second, prior to the main data collection, the preliminary questionnaire was reviewed by a sub-panel of three individuals external to the main Delphi panel: a mother of a preterm infant (end-user perspective), a paediatric nursing faculty member with NICU clinical expertise (clinical perspective), and a digital health engineer (technical perspective). This multi-perspective review was conducted to evaluate item clarity, readability, and the appropriateness of AI–XR terminology and clinical care concepts across stakeholder viewpoints. Minor wording adjustments were made based on the sub-panel's feedback. The final Round 2 instrument evaluated each item across three dimensions: importance, current performance, and future expectations. Although the instrument was developed

**Table 1.** Demographic and professional characteristics of the delphi panel (N = 22).

| Characteristic                    | Categories                               | M $\pm$ SD (range) or n (%)     |
|-----------------------------------|------------------------------------------|---------------------------------|
| Expert Panel (N = 20)             |                                          |                                 |
| Age (year)                        |                                          | 42.35 $\pm$ 6.43 (34–55)        |
| Sex                               | Female                                   | 15 (75)                         |
|                                   | Male                                     | 5 (25)                          |
| Education level                   | Postdoctoral                             | 2 (10)                          |
|                                   | Doctorate                                | 9 (45)                          |
|                                   | Master's (degree awarded)                | 7 (35)                          |
|                                   | Undergraduate                            | 2 (10)                          |
| Specialty field                   | Nurse (NICU nurse or nurse practitioner) | 7 (35)                          |
|                                   | Physician (Professor of Paediatrics)     | 4 (20)                          |
|                                   | Nursing professor                        | 2 (10)                          |
|                                   | Engineer (AI/XR developer)               | 7 (35)                          |
| Years of career                   |                                          | 16.70 $\pm$ 6.25 (9–33)         |
| Caregiver Panel (N = 2)           |                                          |                                 |
| Parental characteristics          |                                          |                                 |
| *Maternal age at delivery (years) |                                          | 33.0 $\pm$ 1.41 (32–34)         |
| Gender                            | Female                                   | 1 (50)                          |
|                                   | Male                                     | 1 (50)                          |
| Job                               | <u>Nurse (on parental leave)</u>         | 1                               |
|                                   | Web designer                             | 1                               |
| Education level                   | Bachelor's                               | 2                               |
| Age                               |                                          | 35.5 $\pm$ 0.71 (35–36)         |
| Multiple birth                    | Yes (Twin)                               | 1 (50)                          |
|                                   | No (Singleton)                           | 1 (50)                          |
| Infant clinical characteristics   |                                          |                                 |
| Gestational age (weeks)           |                                          | 30.1 $\pm$ 2.6 (28.3–32.0)      |
| orrected age (months)             |                                          | 38.5 $\pm$ 14.85 (28–49)        |
| Hospital stays (days)             |                                          | 73.5 $\pm$ 19.09 (60–87)        |
| Birth weight (grams)              |                                          | 1340.0 $\pm$ 317.59 (1140–1770) |
| Birth order                       | First                                    | 1 (50)                          |
|                                   | Second                                   | 1 (50)                          |

\*Maternal age refers to the age of the mother at the time of delivery. For the paternal participant, this value corresponds to the maternal age of his partner at delivery.

AI = artificial intelligence, M  $\pm$  SD = mean  $\pm$  standard deviation, NICU = neonatal intensive care unit, XR = extended reality.

specifically for the Attachment-BRIDGE framework rather than adapted from a pre-existing validated scale, its content validity was rigorously assessed using Lawshe's Content Validity Ratio (CVR), retaining only items meeting the minimum CVR threshold for the given panel size. Internal consistency was additionally confirmed with Cronbach's  $\alpha$  of .91–.94, indicating high reliability across all NICU phases.

### Data analysis

Responses from Round 1 were analysed using qualitative content analysis, including coding, categorisation, and theme clustering, to derive core thematic structures across phases and functional domains. The quantitative data from Round 2 were primarily analysed using IBM SPSS Statistics version 23.0 (IBM Corp., Armonk, NY, USA), following standard Delphi analytic procedures. SPSS was used to compute descriptive statistics (means and standard deviations), CVR based on Lawshe's<sup>26</sup> criteria, consensus and stability indicators (agreement rate, coefficient of variation, and interquartile range), importance–performance gap testing using paired-sample *t*-tests (with Wilcoxon signed-rank tests applied when normality assumptions were violated), expert–caregiver group comparisons (independent-sample *t*-test or Mann–Whitney *U* test), and internal consistency reliability (Cronbach's  $\alpha$ ) for the overall scale, phase-specific domains (E–M–L), and evaluative dimensions. Specific needs-assessment analyses that are not directly supported in SPSS, including the Borich Needs Assessment Index, Importance–Performance Analysis (IPA), and Locus-for-Focus modelling and visualisation, were additionally conducted using R version 4.0.3 (R Foundation for Statistical Computing, Vienna, Austria).

## Ethical considerations

This study was approved by the Institutional Review Board of the Far East University (IRB No. FEUIRB-20250610-01-08) on 10 June 2025. All participants received detailed information regarding the study purpose, procedures, confidentiality, and voluntary participation, and provided written informed consent electronically prior to data collection via an online survey platform (Google Forms). Participation was voluntary, anonymity was ensured, and no personally identifiable information was collected. Detailed ethical procedures, participant protection measures, consent processes, and data management protocols are provided in [Supplementary File S3](#) and illustrated in [Supplementary Figure S2](#) ([Supplementary File S3](#)).

## Results

### Round 1 delphi findings (open-ended survey and interviews)

Qualitative analysis of open-ended responses identified 15 core intervention components that reflect shared priorities and unmet needs in supporting caregivers (parents) of preterm infants during the NICU-to-home transition. These components were organised into three temporal phases to capture stage-specific caregiving needs: early admission (E), middle admission (M), and pre-discharge transition (L). Early-phase priorities focused on emotional stabilisation and adaptation to the NICU environment; middle-phase priorities emphasised parent–infant interaction and attachment formation; and late-phase priorities centred on emergency preparedness and caregiving confidence. Cross-cutting design considerations included immersive AI–XR–based learning, personalised feedback, and usability across hospital and home settings. These findings informed item development for the second round of the Delphi survey ([Supplementary File S4](#)).

### Results of the second delphi round

All 22 participants completed the second round of the survey without missing data. Across all items, importance ratings were consistently higher than performance ratings, which indicated substantial unmet needs. The mean importance score was 4.45 (SD = 0.23), whereas the mean performance score was 2.82 (SD = 0.35), which yielded an average importance–performance gap of 1.63 (SD = 0.31) ([Table 2](#); [Supplementary Table S2](#) in [Supplementary File S2](#)).

Phase-level analysis revealed that the largest gap was during the middle admission phase (mean gap = 1.80), followed by the early admission phase (1.60) and the pre-discharge phase (1.52). This indicated that the caregiver support needs tend to peak during ongoing NICU hospitalisation. Items addressing emotional support, parent–infant interaction, and preparedness for post-discharge care showed high importance but low performance across the phases.

Integrated analyses using IPA, Borich Needs Assessment, and the Locus-for-Focus model consistently identified five highest-priority components: emotional stabilisation (e1), emotional interaction and attachment facilitation (m1), XR-guided kangaroo care (m2), immersive XR-based emotional attunement (m5), and emergency-response simulation (l4) ([Figures 1 and 2](#); [Table 3](#)). These items demonstrated the largest importance–performance gaps and highest Borich scores. Agreement between experts and caregivers was high ( $r = 0.78$ ), supporting the robustness of prioritisation.

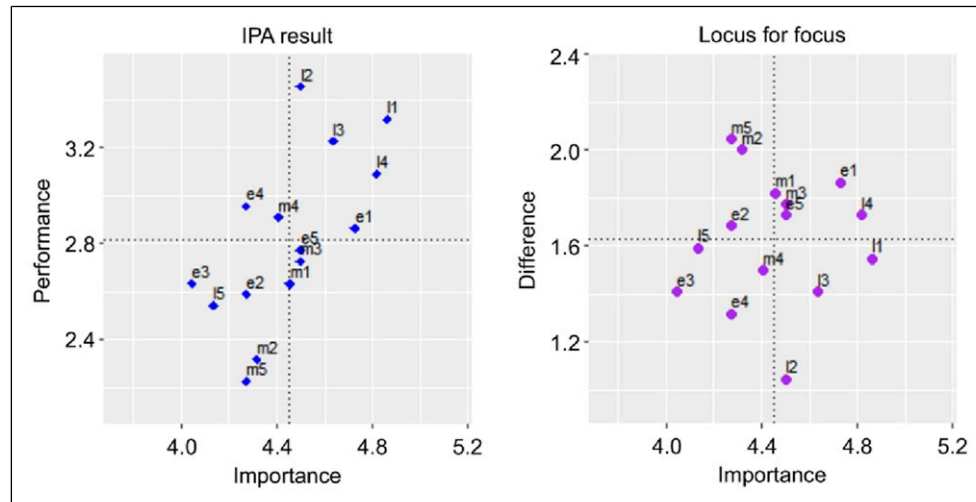
### Validity and reliability of delphi evaluation items

All of the Delphi items met the criteria for content validity and reliability. The CVR values ranged from 0.45 to 0.91, agreement rates exceeded 84% across dimensions, and internal consistency was high (Cronbach's  $\alpha = 0.91$ – $0.94$ ), and these indicated stable measurement across NICU phases.

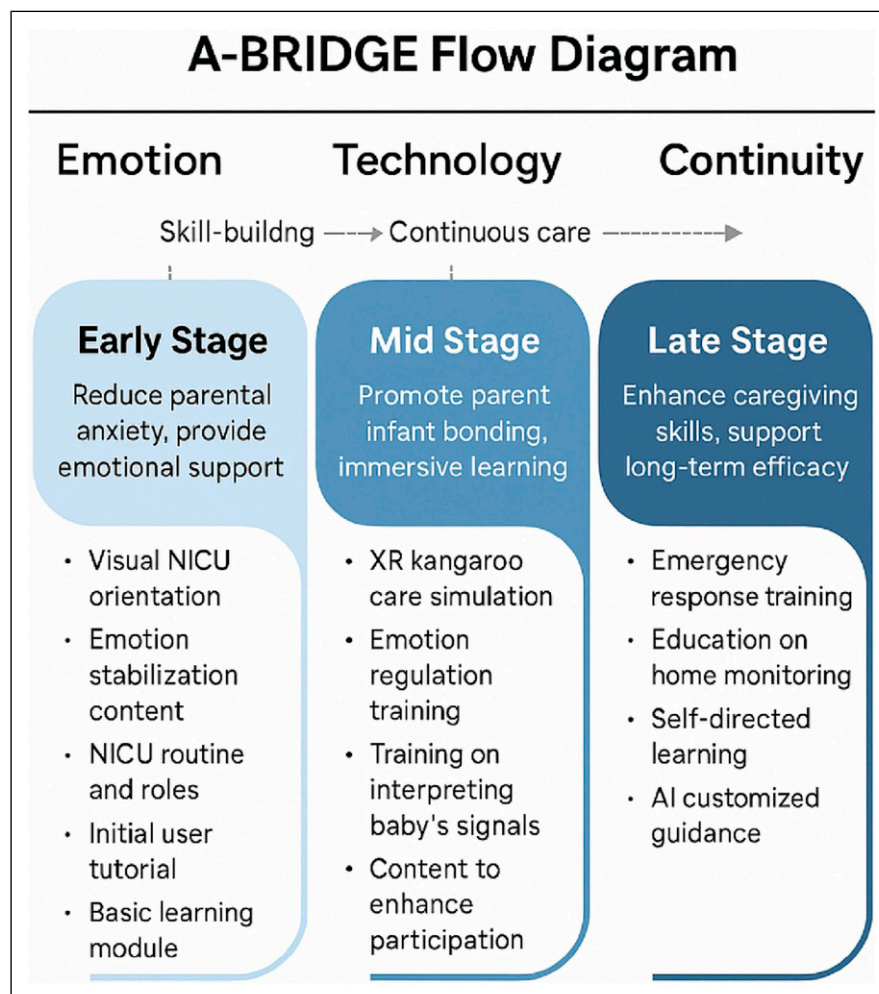
**Table 2.** Importance Performance Analysis Results among participant (N=22).

| Phase      | Importance (mean) | Performance (mean) | Difference (Imp–Perf) | Interpretation                                     |
|------------|-------------------|--------------------|-----------------------|----------------------------------------------------|
| early (e)  | 4.36              | 2.76               | 1.6                   | Need for informational and emotional stabilisation |
| middle (m) | 4.36              | 2.56               | 1.8                   | Highest priority for improvement                   |
| late (l)   | 4.59              | 3.07               | 1.52                  | Basic educational framework already established    |





**Figure 1.** IPA results of the Locus for Focus analysis. IPA = importance–performance analysis.



**Figure 2.** A-BRIDGE stage-based conceptual framework. This figure illustrates the conceptual structure of the A-BRIDGE platform across the NICU-to-home transition phases (early admission, mid-hospitalisation, and pre-discharge). The framework is organised along three core design axes—Emotion, Technology, and Continuity—representing the overarching principles guiding intervention design. AI = artificial intelligence, NICU = neonatal intensive care unit, XR = extended reality.

**Table 3.** Integrated summary of findings (IPA, borich, locus-for-focus).

| Key conclusions                                             | Interpretation                                                                                                            |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Importance > Performance for all items ( $p < .05$ )        | Digital education and XR-based caregiving practices are not yet sufficiently implemented in clinical settings.            |
| Top-priority items = m2 · m5 · e1 · l4                      | Emotional stabilisation, attachment formation, and emergency response represent the core content domains for improvement. |
| Priority by phase = M > E > L                               | The highest need is observed in the “mid-admission emotional/attachment phase”.                                           |
| High agreement between experts and caregivers ( $r = .78$ ) | Indicates strong construct validity and supports the feasibility of real-world clinical application.                      |

IPA = Importance–Performance Analysis.

### Needs assessment for A-BRIDGE content development

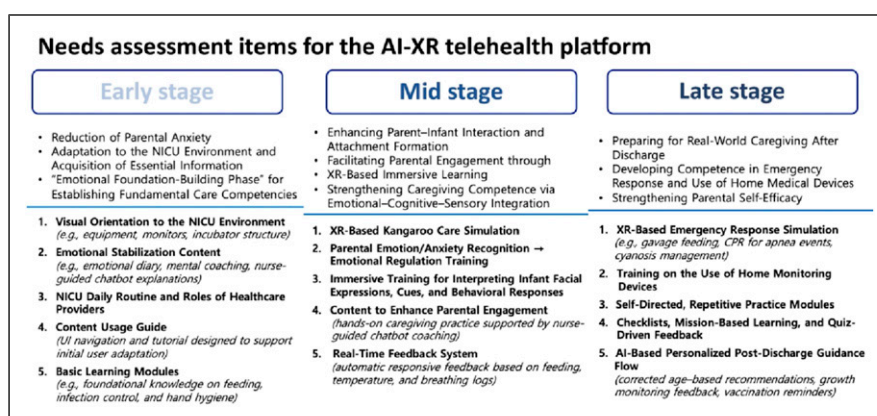
The needs assessment supported a staged A-BRIDGE structure progressing from early emotional stabilisation and environmental adaptation (E), to XR-enabled interaction and attachment formation (M), and finally to emergency preparedness and home-care self-efficacy enhancement (L) (Figure 3). This phase-specific structure provides a validated foundation for subsequent platform development and feasibility testing.

## Discussion

The results of the study indicate a need to focus on four key areas: (1) validation of a time-phased (E–M–L) intervention framework aligned with evolving parental needs; (2) identification of persistent structural gaps in NICU-to-home transitional care; (3) the role of XR-based immersive education in supporting experiential learning; and (4) the contribution of AI-driven personalisation and feedback mechanisms to sustained post-discharge support.

The three-phase structure identified in this study—early admission (E), middle admission (M), and late/pre-discharge transition (L)—is consistent with prior research which demonstrated that parental needs change dynamically across the NICU trajectory.<sup>27–31</sup> During early admission, parents commonly experience emotional shock, fear, and profound uncertainty as they confront their infant’s medical vulnerability and unfamiliar clinical environments. Hua et al.<sup>29</sup> described this period as emotionally destabilising, with parents reporting a sense that their lives had been “turned upside down”, followed by gradual adaptation despite persistent anxiety and guilt. Qualitative studies further indicate that restricted participation and limited information flow during early NICU hospitalisation contribute to feelings of separation and isolation as parents begin to form their parental role.<sup>32</sup> Together, these findings support the importance of early-phase interventions that prioritise emotional stabilisation and environmental orientation, rather than technical skill acquisition alone.

The middle admission phase represents a critical transition during which parents move from being passive observers toward becoming active caregivers. Research suggests that parents begin to experience a sense of authentic parenting only



**Figure 3.** Needs assessment–derived intervention components for the A-BRIDGE platform. This figure presents the 15 intervention components identified through the Delphi-based needs assessment, mapped across the three NICU phases. These components represent the operational elements that inform the implementation of the conceptual framework shown in Figure 2. AI–XR = artificial intelligence–extended reality.

when they are able to engage directly with their infants through interaction and caregiving activities, rather than merely receiving information.<sup>30</sup> FCC research has demonstrated that early and continuous parental involvement during NICU hospitalisation is associated with enhanced parental self-efficacy<sup>33</sup> and stronger parent–infant attachment.<sup>34</sup> Additional evidence indicates that such involvement may mitigate parental anxiety and depressive symptoms over time.<sup>35</sup> This body of literature highlights the middle phase of NICU hospitalisation as a key window for interventions that facilitate emotional connection, active participation, and experiential learning.

During the late or pre-discharge transition phase, parental concerns increasingly focus on readiness for independent caregiving at home. In a qualitative study conducted in Chinese NICUs, caregivers reported that emergency management, medical equipment use, and the applicability of hospital-based knowledge to real-life situations were their primary concerns as discharge approached.<sup>36</sup> Similarly, a multicentre study found that the quality of FCC experiences, communication with healthcare professionals, and opportunities for repeated hands-on training were significant predictors of discharge readiness.<sup>37</sup> Other studies have shown that inadequate discharge planning and fragmented post-discharge support are associated with increased parental anxiety and higher rates of rehospitalisation during the early post-discharge period.<sup>38,39</sup> These findings underscore the importance of continuity-oriented interventions that extend beyond hospital discharge.

Across all phases, sustained mental health and psychosocial support emerges as a cross-cutting need. Although psychological distress frequently begins as acute stress during early NICU admission, a substantial proportion of parents continue to experience anxiety, depression, or posttraumatic stress symptoms after discharge.<sup>40</sup> A scoping review of post-NICU interventions similarly reported persistent emotional distress among parents during the transition home.<sup>28</sup> Swenson et al.<sup>41</sup> further argued that mental health screening and intervention initiated in the NICU should be explicitly linked to community-based services and telehealth follow-up after discharge. These findings reinforce the need for longitudinal mental health support spanning the entire NICU-to-home trajectory.

This study has several limitations that should be acknowledged. First, the Delphi panel was limited to a single national context and a relatively small number of caregiver participants, which may restrict the transferability of our findings to other cultural or healthcare settings. Second, as a consensus-based design study, the findings reflect expert and caregiver priorities rather than empirically tested intervention effects. Third, qualitative inputs were summarised and thematically synthesised, which may have resulted in the loss of some contextual nuance. Fourth, the survey instrument used in this study was developed *de novo* for the Attachment-BRIDGE framework and has not undergone prior longitudinal psychometric validation as a standardised scale. Although content validity was statistically confirmed through CVR-based item screening and item clarity was refined through multi-perspective expert review, the instrument was specifically tailored to the conceptual structure of this platform. Accordingly, the psychometric properties of the finalised instrument should be evaluated in future clinical implementation studies and larger-scale randomised controlled trials. Future research should address these limitations by validating the proposed intervention components through multicentre studies, expanding caregiver representation, and conducting pilot and randomised controlled trials to examine feasibility, acceptability, and mental health outcomes. Integrating longitudinal mental health screening data and real-world usage metrics into future evaluations may further strengthen evidence for AI–XR–based transitional care interventions. Additionally, the A-BRIDGE platform is currently in the prototype development phase, and real-world usability and acceptability data are not yet available. Future studies should evaluate platform usability, ease of navigation, and caregiver experience in both NICU and home settings to establish clinical feasibility and effectiveness in real-world practice.

## Conclusion

This study identified 15 core intervention components for an AI–XR–based telehealth platform (Attachment-BRIDGE), based on consensus from a multidisciplinary panel of 22 participants (CVR: 0.45–0.91; Cronbach's  $\alpha$ : 0.91–0.94). The largest importance–performance gap was observed during the mid-NICU hospitalisation phase (gap = 1.80), indicating the highest unmet caregiver support needs, with emotional stabilisation (e1), attachment facilitation (m1), XR-guided kangaroo care (m2), immersive emotional attunement (m5), and emergency-response simulation (l4) identified as the five top-priority components (expert–caregiver agreement:  $r = 0.78$ ). These findings delineate a three-stage intervention framework (early emotional stabilisation, mid-hospitalisation attachment-focused interaction, and late-phase care preparedness) and provide an evidence-based foundation for designing AI–XR–based telehealth platforms to support families of preterm infants across the NICU-to-home transition.



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## Ethical considerations

This study was approved by the Institutional Review Board of the Far East University (IRB No. FEUIRB-20250610-01-08) on 10 June 2025.

## Consent to participate

All participants received detailed information regarding the study purpose, procedures, confidentiality, and voluntary participation, and provided written informed consent electronically prior to data collection via an online survey platform (Google Forms). Participation was voluntary, anonymity was ensured, and no personally identifiable information was collected.

## Author contributions

**Ah Rim Kim:** Conceptualisation; Methodology; Investigation; Data curation; Formal analysis; Writing – original draft; Writing – review & editing; Supervision; Project administration, Funding Acquisition. **MiJin Choi:** Writing – original draft; Investigation; Data curation; Writing – review & editing. **Jae Eun Sin:** Formal analysis; Software; Visualisation; Writing – review & editing. All authors contributed to the conception and design and/or analysis and interpretation of the data, drafted the manuscript or revised it critically for important intellectual content, approved the final version to be published, and agreed to be accountable for all aspects of the work.

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## Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

## Data Availability Statement

De-identified, aggregated data supporting the findings are available in the Supplementary Files. Additional data may be available from the corresponding author upon reasonable request, subject to ethical approval and participant confidentiality constraints.

## Use of artificial intelligence tools

The authors declare that no generative artificial intelligence (AI) tools were used in the development of the scientific content of this manuscript. AI-assisted tools were used solely for language editing and formatting purposes, and all intellectual content, analysis, and interpretations were generated and verified by the authors.

## Supplemental material

Supplemental material for this article is available online.

## References

1. OECD. *Society at a glance 2024: OECD social indicators*. OECD Publishing, 2024. <https://doi.org/10.1787/918d8db3-en> (Accessed 10 February 2026).
2. Awoyesuku DPA, Okagua DKE, Iwo-Amah DRS, et al. Pregnancy outcomes in women of advanced maternal age: A retrospective cohort study. *Int J Clin Obstet Gynaecol* 2024; 8: 20–28. <https://doi.org/10.33545/gynae.2024.v8.i2a.1430>
3. United Nations Department of Economic and Social Affairs Population Division. *World population prospects 2024: Summary of results*. United Nations, 2024. [https://population.un.org/wpp/assets/Files/WPP2024\\_Summary-of-Results.pdf](https://population.un.org/wpp/assets/Files/WPP2024_Summary-of-Results.pdf) (Accessed 10 February 2026).
4. Statistics Korea. *Birth Statistics in 2024*. Daejeon: Statistics Korea. <https://kosis.kr> (2025), Accessed 5 February 2026.

5. World Health Organization. *Preterm birth*, 2023. <https://www.who.int/news-room/fact-sheets/detail/preterm-birth> (Accessed 5 February 2026).
6. Provenzi L, Khashu M, McLean M, et al. Editorial: Psychological implications of preterm birth. *Front Psychol* 2025; 16: 1572801. <https://doi.org/10.3389/fpsyg.2025.1572801>
7. Song IG, Lim JW, Chang YS, et al. Improved outcomes of very low birth weight infants in Korea: 2015–2016 vs. 2021–2022 from the Korean Neonatal Network. *J Korean Med Sci* 2025; 40: e258. <https://doi.org/10.3346/jkms.2025.40.e258>
8. Yoon S, Park J, Lee H, et al. Influence of partnerships with nurses and social support on readiness for discharge among mothers of premature infants. *Child Health Nurs Res* 2019; 25: 417–424. <https://doi.org/10.4094/chnr.2019.25.4.417>
9. Jung SY and Tak YR. Family-centered care for hospitalized children: Concept analysis. *Child Health Nurs Res* 2017; 23: 28–36. <https://doi.org/10.4094/chnr.2017.23.1.28>
10. Lee J. Neonatal family-centered care: Evidence and practice models. *Clin Exp Pediatr* 2023; 67: 171–177. <https://doi.org/10.3345/cep.2023.00367>
11. Al-Motlaq MA, Carter B, Neill S, et al. Toward developing consensus on family-centred care: An international descriptive study and discussion. *J Child Health Care* 2019; 23: 458–467. <https://doi.org/10.1177/1367493518795341>
12. Williams L. Impact of family-centered care on pediatric and neonatal intensive care outcomes. *AACN Adv Crit Care* 2016; 27: 158–161. <https://doi.org/10.4037/aacnacc2016579>
13. Bellizzi S, Panu Napodano CMP and Murgia P. Family-centered care for newborns: A global perspective and review. *J Trop Pediatr* 2024; 70: fmae026. <https://doi.org/10.1093/tropej/fmae026>
14. Lim S-J and Bang K-S. The perceptions and performance of family-centered care among pediatric nurses at a Children's Hospital in South Korea: A descriptive study. *Child Health Nurs Res* 2023; 29: 207–217. <https://doi.org/10.4094/chnr.2023.29.3.207>
15. Jung SY. Health care providers' perceptions of family-centered care in pediatrics. *Child Health Nurs Res* 2018; 24: 465–474. <https://doi.org/10.4094/chnr.2018.24.4.465>
16. Yang X, Zhang J, Cao M, et al. Application of e-health on neonatal intensive care unit discharged preterm infants and their parents: Protocol for systematic review and meta-analysis. *Digit Health* 2023; 9: 20552076231205271. <https://doi.org/10.1177/20552076231205271>
17. Wagenaar J, Mah C, Bodell F, et al. Opportunities for telemedicine to improve parents' well-being during the neonatal care journey: Scoping review. *JMIR Pediatr Parent* 2024; 7: e60610. <https://doi.org/10.2196/60610>
18. Wasswa W and Ware JA. A systematic literature review on virtual reality applications in medical education. *AETIC* 2025; 9: 50–68. <https://doi.org/10.33166/aetic.2025.03.004>
19. Ataguba G, Aworinde H, Anukem S, et al. Systematic review of AR/VR applications for health and wellness: Trends and future opportunities. *Int J Hum-Comput Interact* 2025; 1–45. <https://doi.org/10.1080/10447318.2025.2588676>
20. Gordon TJ and Helmer O. *Report on a long-range forecasting study*. Rand Corporation, 1964. <https://www.rand.org/pubs/papers/P2982.html> (Accessed 1 February 2026).
21. Niederberger M, Schifano J, Deckert S, et al. Delphi studies in social and health sciences—Recommendations for an interdisciplinary standardized reporting (DELPHISTAR): Results of a Delphi study. *PLOS One* 2024; 19: e0304651. <https://doi.org/10.1371/journal.pone.0304651>
22. Ziglio E. The Delphi method and its contribution to decision-making. In: Adler M and Ziglio E (eds). *Gazing into the oracle: The Delphi method and its application to social policy and public health*. Jessica Kingsley Publishers, 1996, pp. 3–33.
23. Skulmoski GJ, Hartman FT and Krahn J. *The Delphi method for graduate research*. : JITE, 2007, vol 6, pp. 1–21.
24. Kim DE, Sagong H and Yoon JY. A Delphi study for developing a person-centered dementia care online education program in long-term care facilities. *J Korean Acad Community Health Nurs* 2019; 30: 295–306. <https://doi.org/10.12799/jkachn.2019.30.3.295>
25. Kim C, Han SY and Chin YR. A Delphi study for development of disaster nursing education contents in community health nursing. *J Korean Acad Community Health Nurs* 2021; 32: 555–565. <https://doi.org/10.12799/jkachn.2021.32.4.555>
26. Lawshe CH. A quantitative approach to content validity. *Pers Psychol* 1975; 28: 563–575. <https://doi.org/10.1111/j.1744-6570.1975.tb01393.x>
27. Kim AR. Addressing the needs of mothers with infants in the neonatal intensive care unit: A qualitative secondary analysis. *Asian Nurs Res* 2020; 14: 327–337. <https://doi.org/10.1016/j.anr.2020.09.004>
28. Griffith T, Singh A, Naber M, et al. Scoping review of interventions to support families with preterm infants post-NICU discharge. *J Pediatr Nurs* 2022; 67: e135–e149. <https://doi.org/10.1016/j.pedn.2022.08.014>
29. Hua W, Zhou J, Wang L, et al. “It turned my life upside down”: Parents' emotional experience of the transition with their preterm infant from birth to discharge home—A qualitative study. *Aust Crit Care* 2023; 36: 679–686. <https://doi.org/10.1016/j.aucc.2022.10.007>
30. Lisle J, Buma K, Smith J, et al. Maternal perceptions about sensory interventions in the neonatal intensive care unit: An exploratory qualitative study. *Front Pediatr* 2022; 10: 884329. <https://doi.org/10.3389/fped.2022.884329>
31. Adama EA, Adua E, Bayes S, et al. Support needs of parents in neonatal intensive care unit: An integrative review. *J Clin Nurs* 2022; 31: 532–547. <https://doi.org/10.1111/jocn.15972>

32. Kynø NM, Fugelseth D, Knudsen LMM, et al. Starting parenting in isolation a qualitative user-initiated study of parents' experiences with hospitalization in neonatal intensive care units during the COVID-19 pandemic. *PLOS One* 2021; 16: e0258358. <https://doi.org/10.1371/journal.pone.0258358>
33. Pineda R, Bender J, Hall B, et al. Parent participation in the neonatal intensive care unit: Predictors and relationships to neurobehavior and developmental outcomes. *Early Hum Dev* 2018; 117: 32–38. <https://doi.org/10.1016/j.earlhumdev.2017.12.008>
34. Oude Maatman SM, Bohlin K, Lilliesköld S, et al. Factors influencing implementation of family-centered care in a neonatal intensive care unit. *Front Pediatr* 2020; 8: 222. <https://doi.org/10.3389/fped.2020.00222>
35. Velasco Arias JM, Peres AM, Escandell Rico FM, et al. Developmental-centered care in preterm newborns: Scoping review. *Children (Basel)* 2025; 12: 783. <https://doi.org/10.3390/children12060783>
36. Ma RH, Zhang Q, Ni ZH, et al. Transitional care experiences of caregivers of preterm infants hospitalized in a neonatal intensive care unit: A qualitative descriptive study. *Nurs Open* 2021; 8: 3484–3494. <https://doi.org/10.1002/nop2.899>
37. Franck LS, Kriz RM, Bisgaard R, et al. Parent readiness for their preterm infant's neonatal intensive care unit discharge. *J Perinat Neonatal Nurs* 2023; 37: 68–76. <https://doi.org/10.1097/JPN.0000000000000612>
38. Enlow E, Herbert SL, Jovel IJ, et al. Neonatal intensive care unit to home: The transition from parent and pediatrician perspectives, a prospective cohort study. *J Perinatol* 2014; 34: 761–766. <https://doi.org/10.1038/jp.2014.75>
39. Boykova M and Kenner C. Transition from hospital to home for parents of preterm infants. *J Perinat Neonatal Nurs* 2012; 26: 81–87, quiz 88. <https://doi.org/10.1097/JPN.0b013e318243e948>
40. Malliarou M, Karadonta A, Mitroulas S, et al. Preterm parents' stress and coping strategies in a Neonatal Intensive Care Unit in a University Hospital of Central Greece. *Mater Sociomed* 2021; 33: 244–249. <https://doi.org/10.5455/msm.2021.33.244-249>
41. Swenson SA, Desai RK, Velagala S, et al. From NICU to home: Meeting the mental health needs of families after discharge. *J Perinatol* 2025; 45: 1863–1870. <https://doi.org/10.1038/s41372-025-02503-x>